

Model Selection

CS1090A Introduction to Data Science

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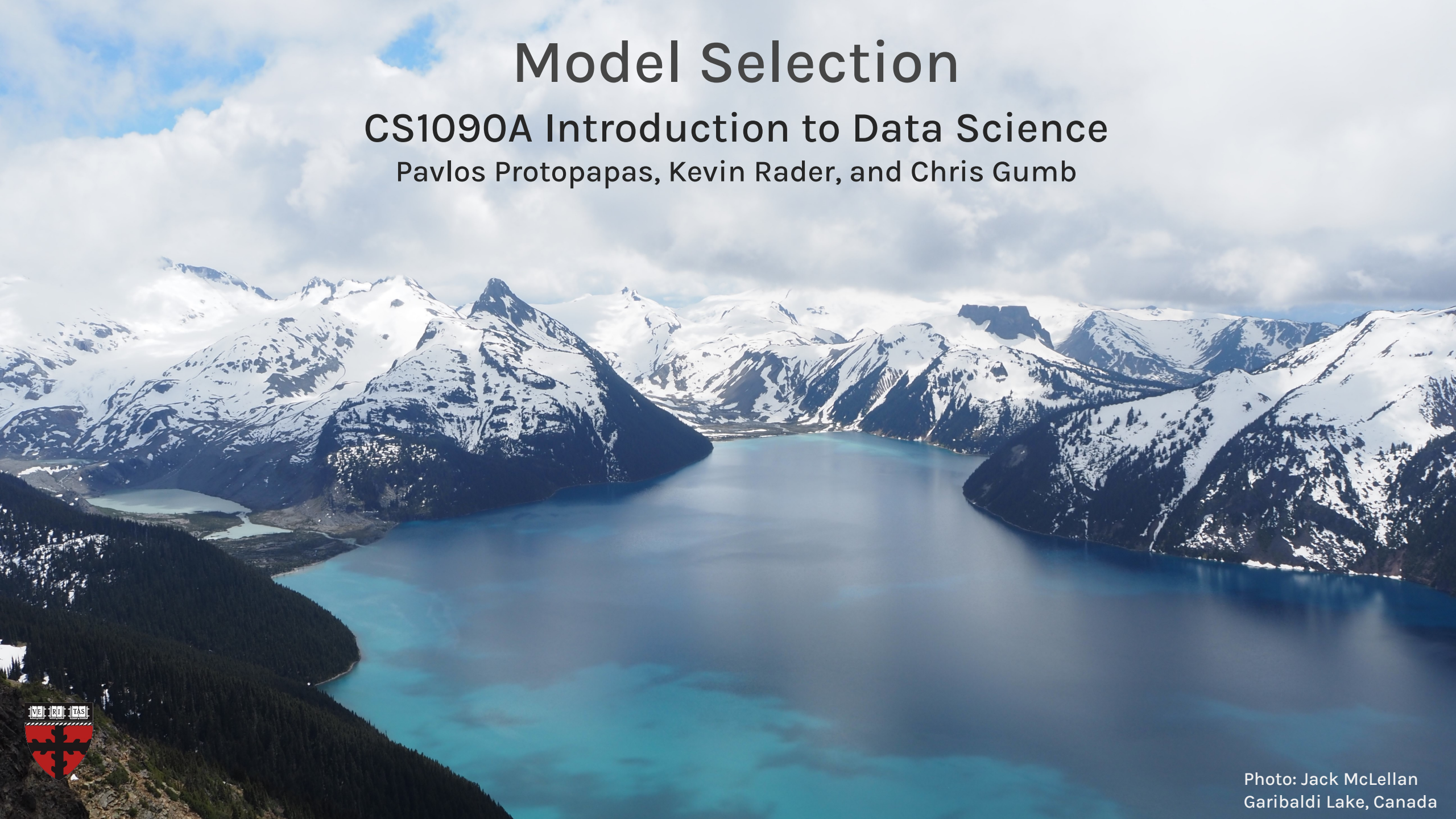


Photo: Jack McLellan
Garibaldi Lake, Canada

Lecture Outline

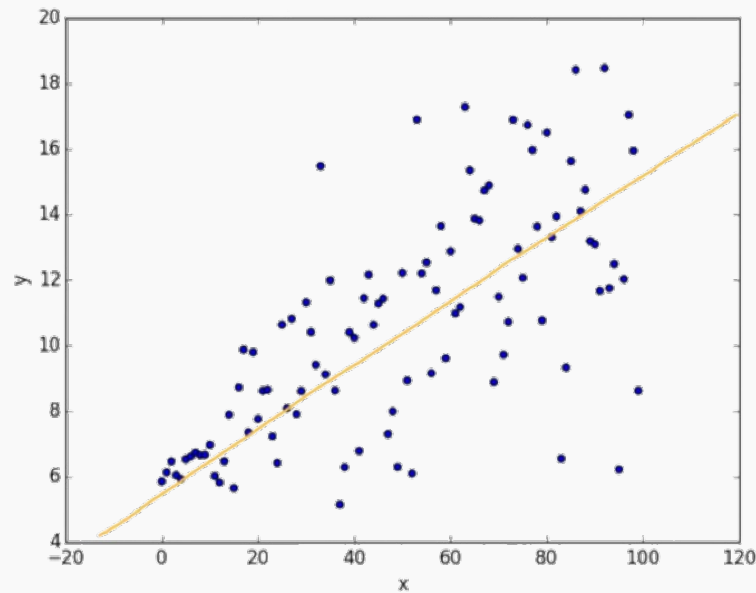
Interaction Effects in Regression Models

Polynomial Regression: Extending Linear Models

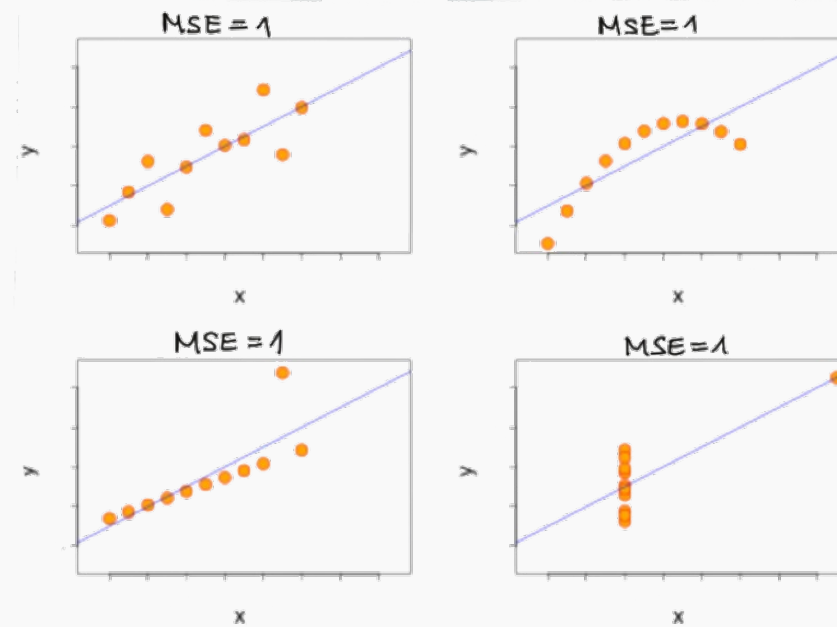
Model Selection Techniques: Focus on Cross-Validation

Evaluation: Training Error

Just because we found the model that minimizes the squared error it doesn't mean that it's a good model. We could investigate the R^2 but also:



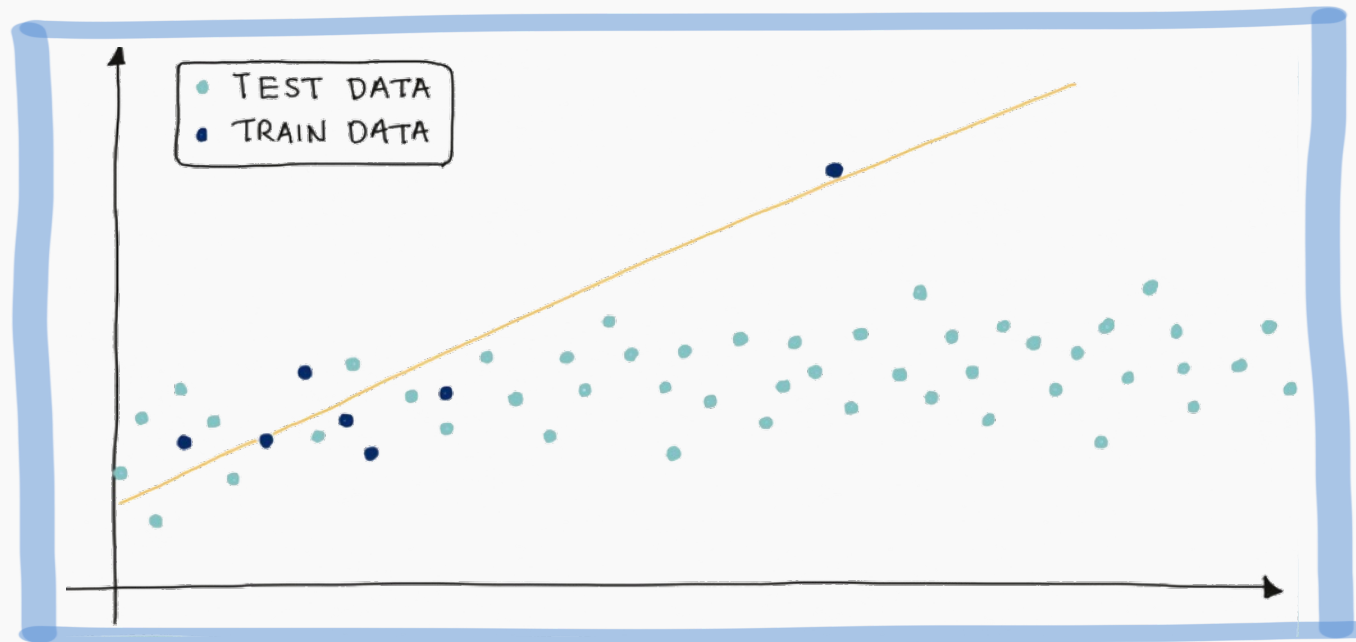
The MSE is high due to noise in the data.



The MSE is high in all four models, but the models are not equal.

Evaluation: Test Error

We need to evaluate the fitted model on new data, data that the model did not train on, the **unseen data**.



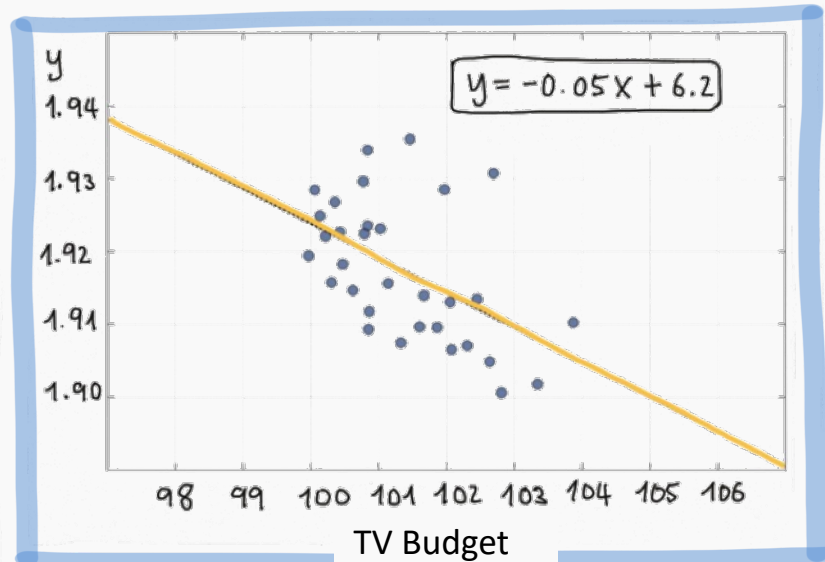
The **training** MSE here is 2.0 where the **test** MSE is 12.3.

The training data contains a strange point – an outlier – which confuses the model.

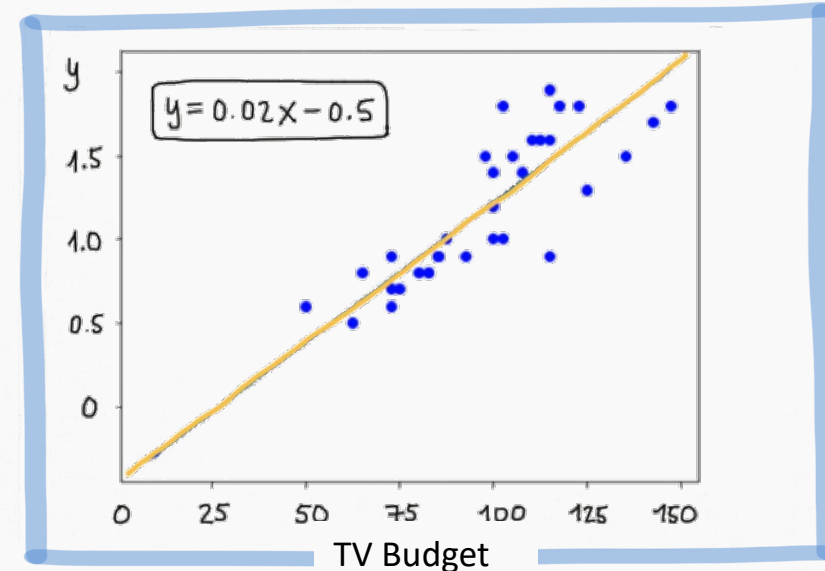
Fitting to meaningless patterns in the training is called **overfitting**.

Evaluation: Model Interpretation

For linear models it's important to interpret the parameters



The MSE of this model is very small. But the slope is -0.05. That means the larger the budget the less the sales.



The MSE is very small, but the intercept is -0.5 which means that for very small budget we will have negative sales.

Generalization Error

We know to evaluate the model on both train and unseen data, because models that do well on training data may do poorly on new data (overfitting).

The ability of models to do well on new data is called **generalization**.

The goal of **model selection** is to choose the model that generalizes the best.

Model Selection

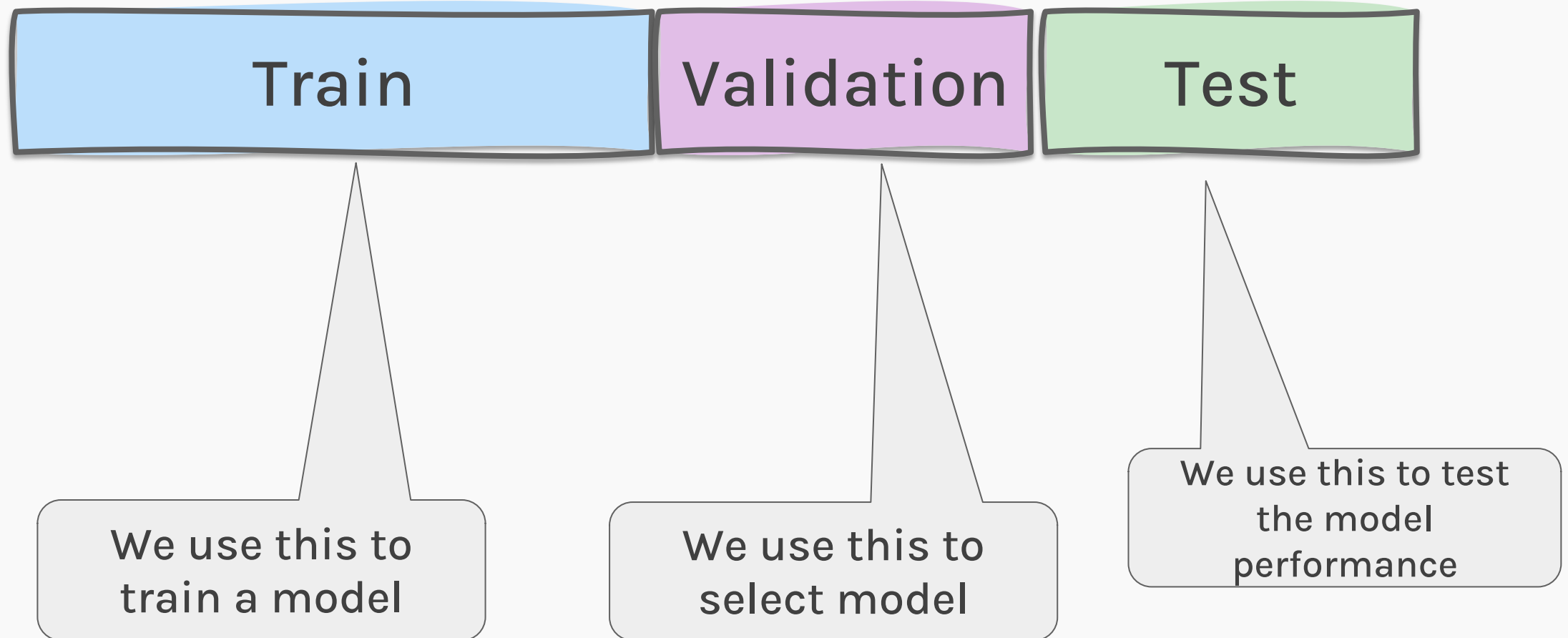
Model selection is the application of a principled method to determine the complexity of the model, e.g., choosing a subset of predictors, choosing the degree of the polynomial model etc.

A strong motivation for performing model selection is to avoid **overfitting**, which we saw can happen when:

- there are too many predictors:
 - the feature space has high dimensionality
 - the polynomial degree is too high
 - too many cross terms are considered
- the coefficients values are too **extreme** (we have not seen this yet)

Train-Validation-Test

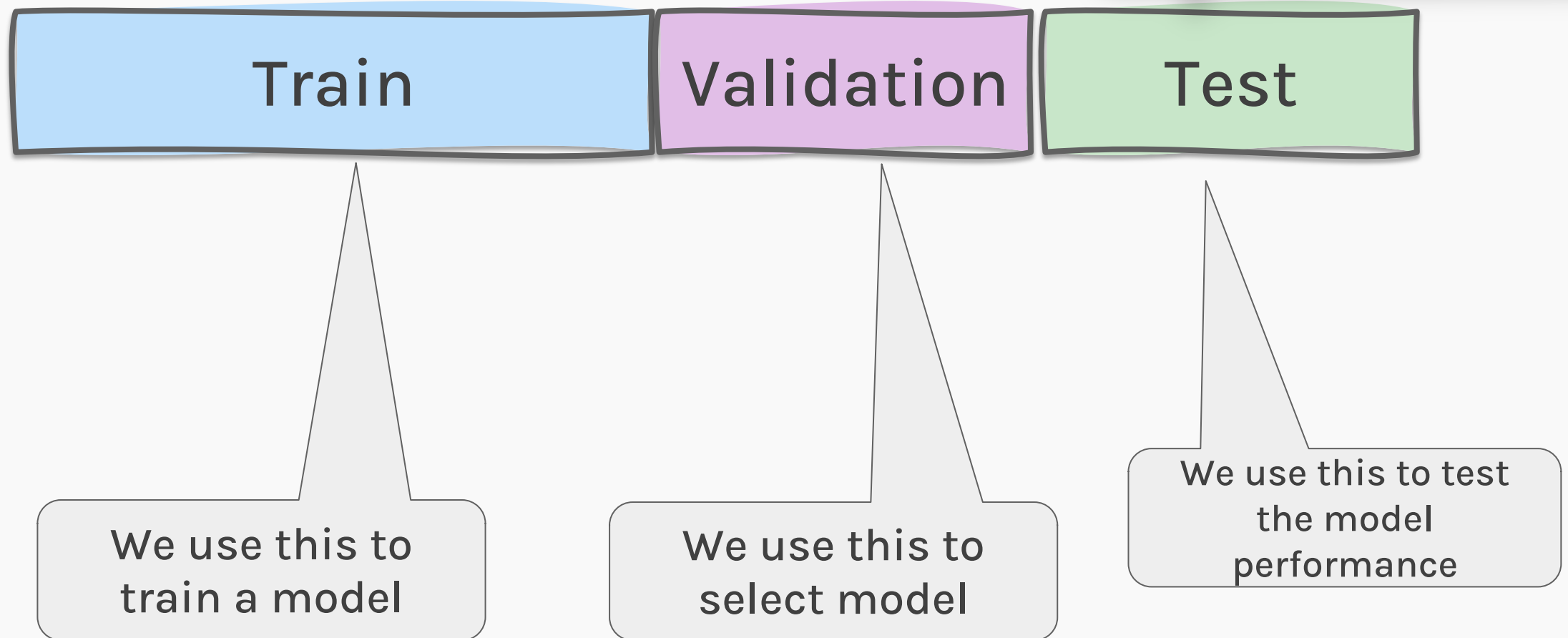
As we all know, we have 3 splits of the data split



Train-Validation-Test

We introduce a different sub-set, which we called validation to select the model.

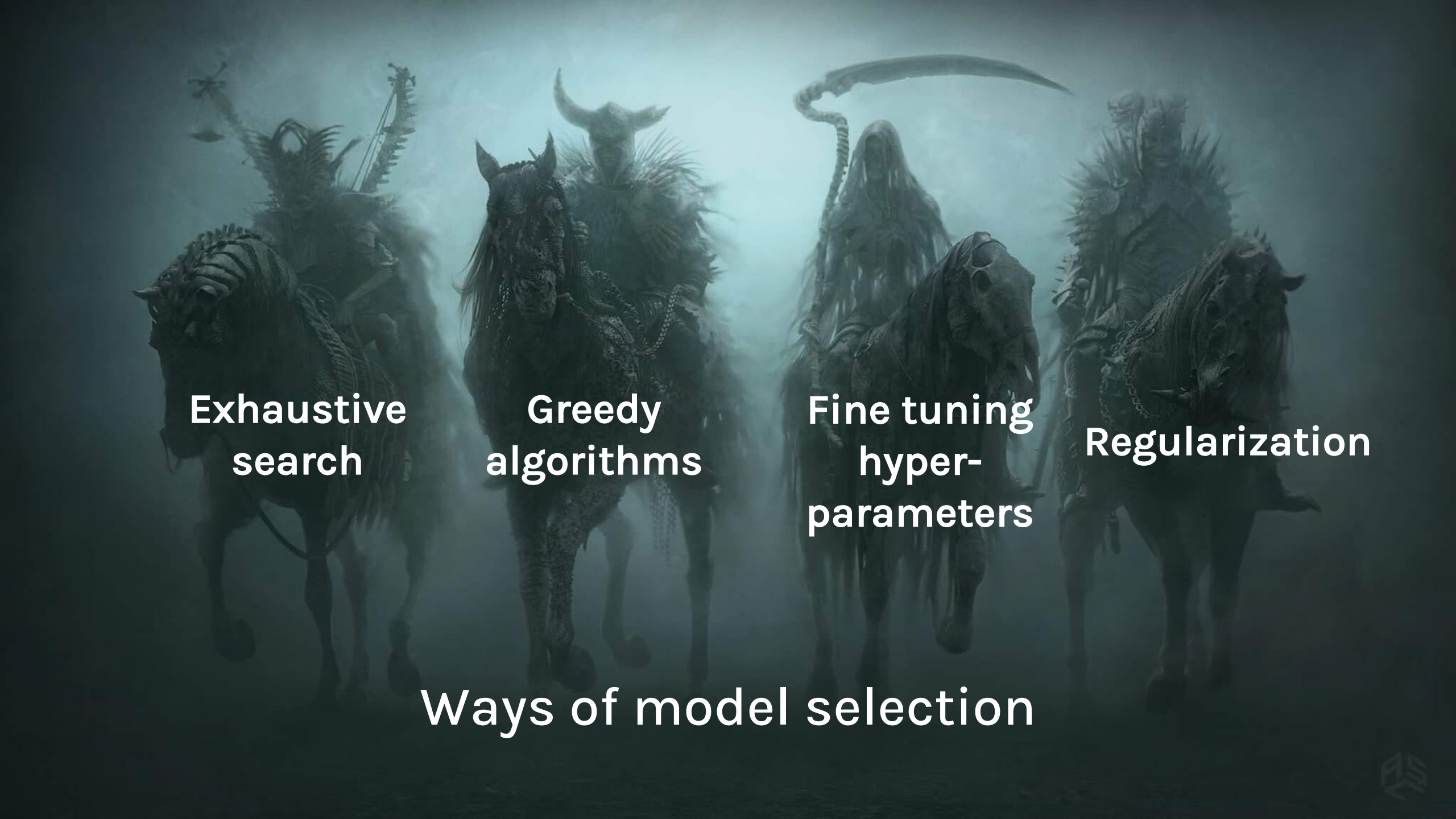
The test set should never be touched for model training or selection.



Model Selection

Ways of model selection:

- Exhaustive search
- Greedy algorithms
- Fine tuning hyper-parameters
- Regularization

The image features four spectral horses galloping towards the viewer in a dark, misty environment. Each horse is ridden by a spectral figure. The first horse on the left is ridden by a figure holding a pair of scales. The second horse is ridden by a figure with a bull's head. The third horse is ridden by a figure holding a scythe. The fourth horse is ridden by a figure with a crown. The text labels are overlaid on the image, and the title is at the bottom.

**Exhaustive
search**

**Greedy
algorithms**

**Fine tuning
hyper-
parameters**

Regularization

Ways of model selection



Model Selection

Ways of model selection:

- **Exhaustive search**
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Model Selection: How many models?

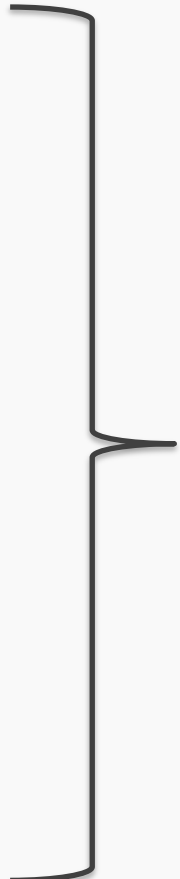
Can you prove this?

Question:

How many different models can we form when considering J predictors (with only linear terms)?

Example: 3 predictors (X_1, X_2, X_3)

- Models with 0 predictor:
M0:
- Models with 1 predictor:
M1: X_1
M2: X_2
M3: X_3
- Models with 2 predictors:
M4: $\{X_1, X_2\}$
M5: $\{X_2, X_3\}$
M6: $\{X_3, X_1\}$
- Models with 3 predictors:
M7: $\{X_1, X_2, X_3\}$



2^J models

Model Selection

Ways of model selection:

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- **Greedy algorithms**
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Stepwise Variable Selection and Validation

Selecting optimal subsets of predictors (including choosing the degree of polynomial models) through:

- stepwise variable selection - **iteratively building** an optimal subset of **predictors** by optimizing a fixed model evaluation metric each time.
- **selecting** an optimal **model** by evaluating each model on validation set.

Stepwise Variable Selection: Forward method

In forward selection, we find an ‘optimal’ set of predictors by iterative building up our set.

1. Start Simple

Begin with the empty set P_0 , construct the null model M_0

2. Add Predictors One by One

For $k = 1, \dots, J$:

2.1 Use the best model so far, M_{k-1} , based on $k - 1$ predictors

2.2 Pick one predictor not yet P_{k-1} that improves the model the most according to validation MSE

2.3 Create a new model, M_k , using the selected predictors

3. Select the model M amongst $\{M_0, M_1, \dots, M_J\}$ that optimizes validation MSE

Stepwise Variable Selection Computational Complexity

How many models did we evaluate?

- 1st step, J Models
- 2nd step, $J-1$ Models (add 1 predictor out of $J-1$ possible)
- 3rd step, $J-2$ Models (add 1 predictor out of $J-2$ possible)
- ...

$$O(J^2) \ll 2^J \text{ for large } J$$

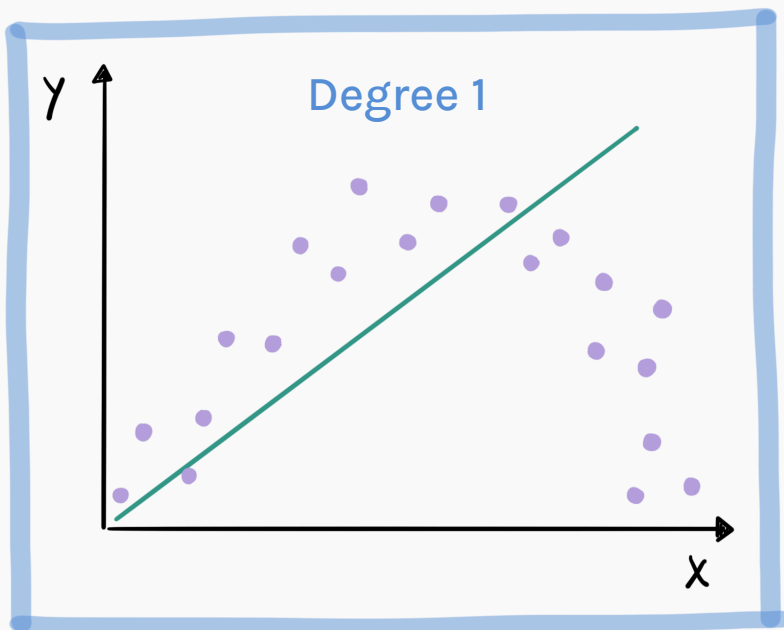
Model Selection

Ways of model selection:

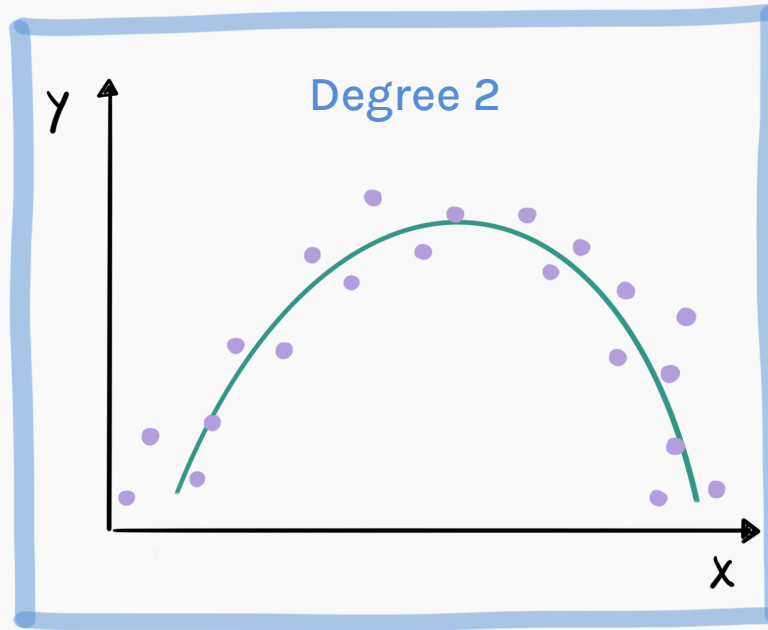
- Exhaustive search
- Greedy algorithms
- **Fine tuning hyper-parameters**
- Regularization

Choosing the degree of the polynomial model

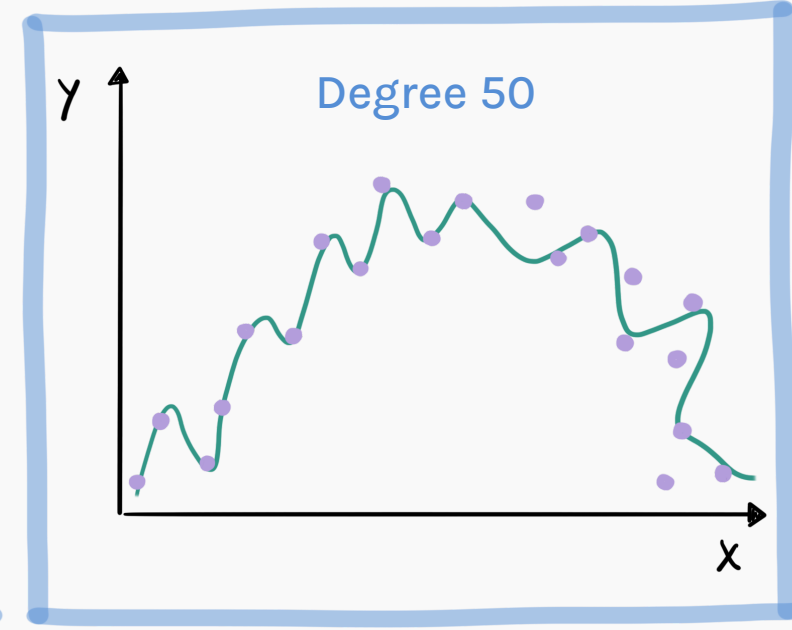
We turn model selection into choosing a **hyper-parameter**. For example, polynomial regression requires choosing a degree – this can be thought as model selection – and we select the model by tuning the hyper-parameter.



Underfitting: when the degree is too low, the model cannot fit the trend.



We want a model that fits the trend and ignores the noise.



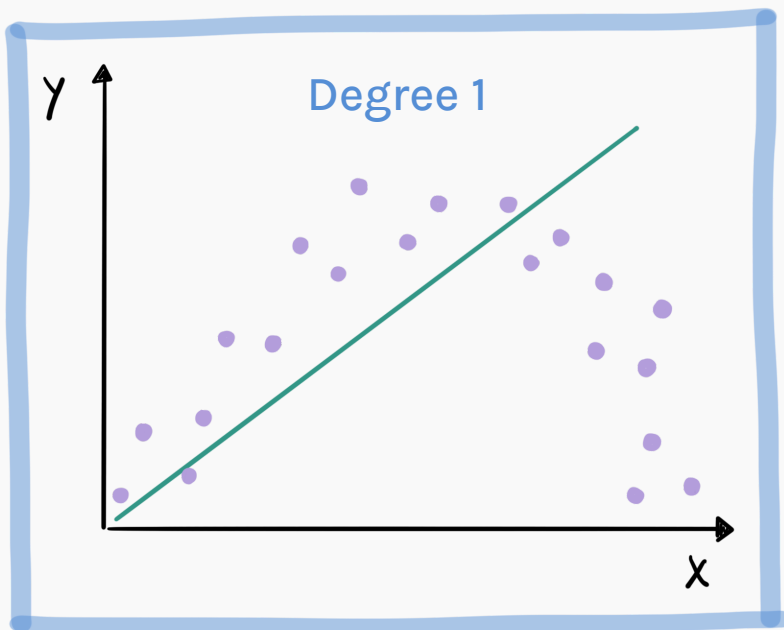
Overfitting: when the degree is too high, the model fits all the noisy data points.

Choosing the degree of the polynomial model

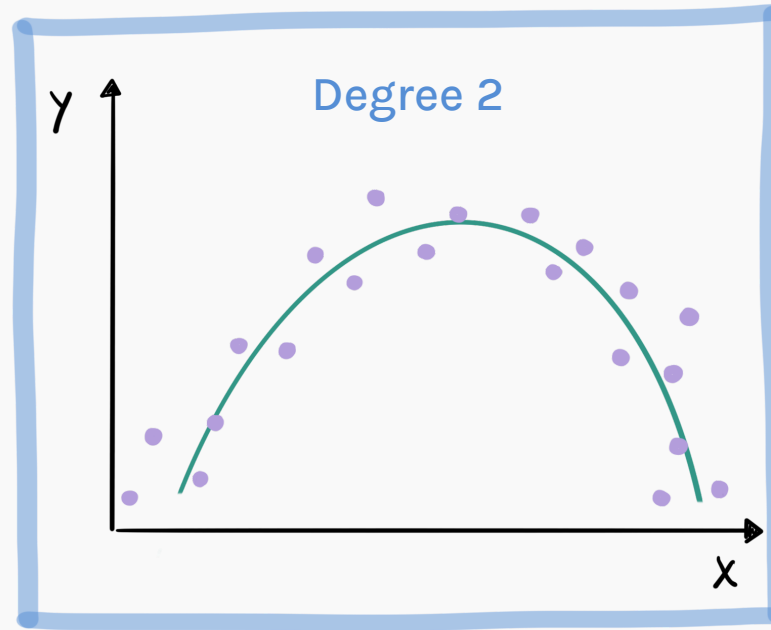
kNN: k was a hyper-parameter



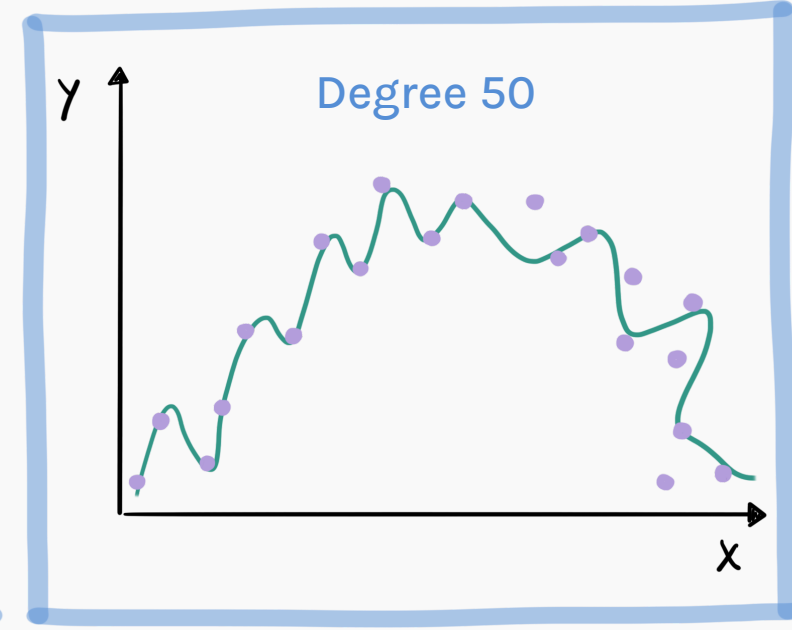
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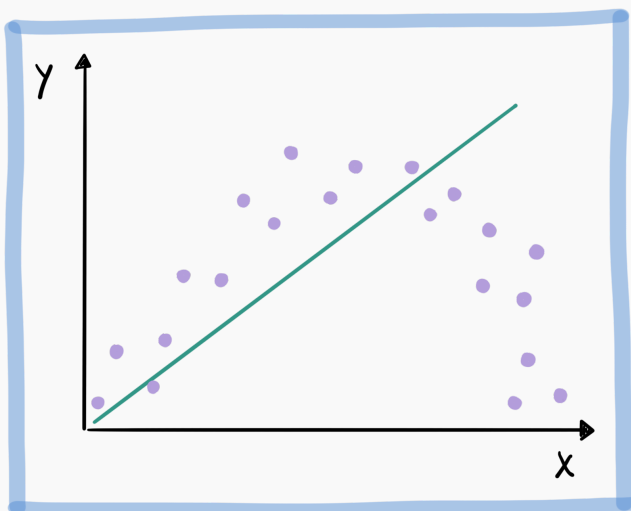


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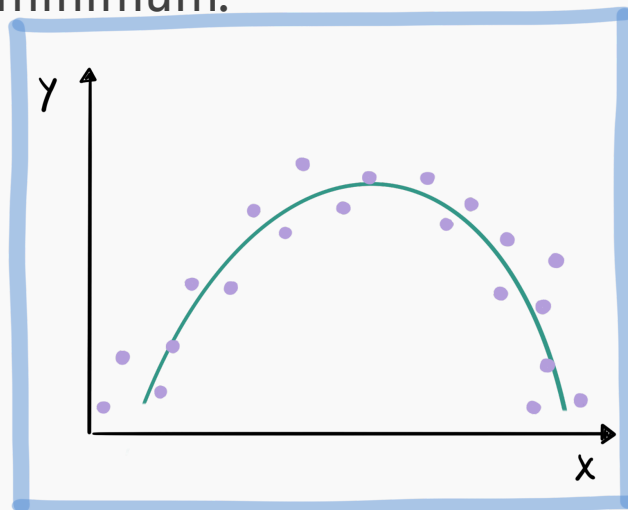


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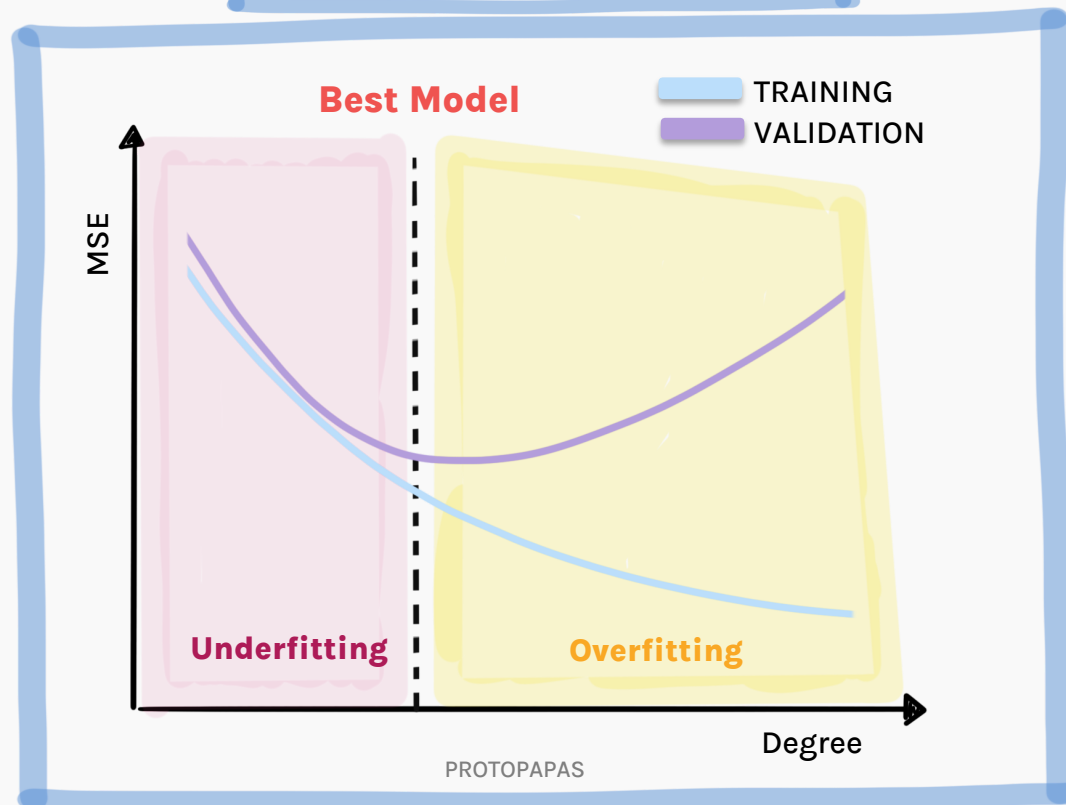
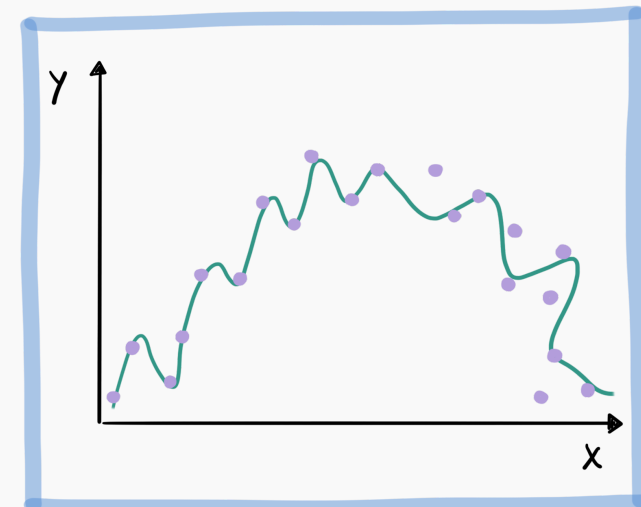
Underfitting: train and validation error is high.



Best model: validation error is minimum.



Overfitting: train error is low, validation error is high.



What are the parameters of the models and what are the hyperparameters?